Coverage Metric - GPU Optimized

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Table of contents


```python
# Configuración para visualización en PDF
import pandas as pd
import numpy as np
import os

pd.set_option('display.max_colwidth', None)
pd.set_option('display.max_columns', None)
pd.set_option('display.width', 100)
np.set_printoptions(linewidth=100, edgeitems=3)

os.environ['COLUMNS'] = '100'

print(" Configuración para PDF lista")
```

```
 Configuración para PDF lista
```

```python
# Metadata para naming de checkpoints (separada de hiperparámetros)
NAMING_META = {
    'variante': 'topk',          # Arquitectura del SAE
    'tecnica': 'batch-topk',                 # L1 con penalty annealing
    'capa': 6,                       # Layer del modelo OthelloGPT
    'juegos': 1500,                  # Número de partidas
    'base_path': 'C:\\Users\\Esposa\\Documents\\Repos\\sae-othello-gpt',  # Path base para checkpoints
    'experiment_base_path': os.path.abspath('../../experiments') ,      # sae/experiments/
    'extras': {
       'expansion': 8,   # d_sae / d_in = 8192 / 512
        'k': 50,           # número de features activas L0 promedio   k
        'epochs': 1000,
        'patience': 20
    }
}
print('\n Metadata del experimento:')
for key, value in NAMING_META.items():
    print(f'  {key}: {value}')
```

```
Metadata del experimento:
 variante: topk
 tecnica: batch-topk
 capa: 6
 juegos: 1500
 base_path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt
 experiment_base_path: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments
 extras: {'expansion': 8, 'k': 50, 'epochs': 1000, 'patience': 20}
```

0.1 1. Setup e Imports

```python
import numpy as np
import torch
import torch.nn as nn
import torch.nn.functional as F
import sys
import os
from pathlib import Path
from tqdm import tqdm
import time

# Configurar paths
project_root = Path('../../..').resolve()
sys.path.insert(0, str(project_root))

from sae.tools.naming_utils import get_checkpoint_dir, get_model_name, get_extras_id
from sae.tools.experiment_utils import get_experiment_dir, get_metrics_dir, get_report_name

# Verificar GPU
device = torch.device('cuda' if torch.cuda.is_available() else 'cpu')
print(f"Device: {device}")
if torch.cuda.is_available():
    print(f"GPU: {torch.cuda.get_device_name(0)}")
    print(f"Memoria disponible: {torch.cuda.get_device_properties(0).total_memory / 1e9:.2f} GB")
```

```
Device: cuda
GPU: NVIDIA GeForce RTX 5060
Memoria disponible: 8.55 GB
```

```python
class BatchTopKSAE(nn.Module):
    def __init__(self, d_in: int, d_sae: int, k: int, k_aux: int = 512):
        super().__init__()
        self.d_in  = d_in
        self.d_sae = d_sae
        self.k     = k
        self.k_aux = k_aux

        self.W_enc = nn.Parameter(torch.empty(d_in, d_sae))
        self.W_dec = nn.Parameter(torch.empty(d_sae, d_in))
        self.b_dec = nn.Parameter(torch.zeros(d_in))

        nn.init.kaiming_uniform_(self.W_enc)
        nn.init.kaiming_uniform_(self.W_dec)
        self.W_dec.data = self.W_dec.data / self.W_dec.data.norm(dim=-1, keepdim=True)
        self.register_buffer('num_batches_not_active', torch.zeros(d_sae))

    def encode(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
        x_norm, _, _ = self.preprocess_input(x)
        x_cent   = x_norm - self.b_dec
```

```python
        pre_acts = F.relu(x_cent @ self.W_enc)
        return self._topk_activation(pre_acts, threshold=threshold)

    def _topk_activation(self, x: torch.Tensor, threshold=None) -> torch.Tensor:
        if threshold is None:
            acts_topk = torch.topk(x.flatten(), self.k * x.shape[0], dim=-1)
            return (
                torch.zeros_like(x.flatten())
                .scatter(-1, acts_topk.indices, acts_topk.values)
                .reshape(x.shape)
            )
        return torch.where(x > threshold, x, torch.zeros_like(x))

    def decode(self, hidden: torch.Tensor) -> torch.Tensor:
        return hidden @ self.W_dec + self.b_dec

    def preprocess_input(self, x: torch.Tensor):
        x_mean = x.mean(dim=-1, keepdim=True)
        x = x - x_mean
        x_std = x.std(dim=-1, keepdim=True)
        x = x / (x_std + 1e-5)
        return x, x_mean, x_std

    def postprocess_output(self, x_reconstruct, x_mean, x_std):
        return x_reconstruct * x_std + x_mean

    def forward(self, x: torch.Tensor, threshold=None):
        x_norm, x_mean, x_std = self.preprocess_input(x)
        x_cent   = x_norm - self.b_dec
        pre_acts = F.relu(x_cent @ self.W_enc)
        hidden   = self._topk_activation(pre_acts, threshold=threshold)
        recon    = self.postprocess_output(self.decode(hidden), x_mean, x_std)
        return recon, hidden, pre_acts

print("BatchTopKSAE definida")
```

```
BatchTopKSAE definida
```

0.2 2. Cargar Datos

```python
# Cargar activaciones pre-SAE
activations_path = project_root / "sae" / "activations" / "data" / "layer6_2000games.npy"
activations_full = np.load(activations_path)
activations = activations_full

print(f"Activaciones del modelo:")
print(f"  Shape original: {activations_full.shape}")
print(f"  Shape final: {activations.shape}")
print(f"  Memoria: {activations.nbytes / (1024**2):.2f} MB")
```

```
Activaciones del modelo:
  Shape original: (118000, 512)
  Shape final: (118000, 512)
  Memoria: 230.47 MB
```

```python
# Cargar ground truth de BSPs
bsp_gt_path = project_root / "sae" / "metrics" / "02_data" / "bsp_ground_truth_2000games.npy"
bsp_names_path = project_root / "sae" / "metrics" / "02_data" / "bsp_ground_truth_2000games.names.npy"
```

```python
bsp_ground_truth_full = np.load(bsp_gt_path)
bsp_names = np.load(bsp_names_path, allow_pickle=True)
bsp_ground_truth = bsp_ground_truth_full
print(f"\nGround truth BSPs:")
print(f"  Shape original: {bsp_ground_truth_full.shape}")
print(f"  Shape final: {bsp_ground_truth.shape}")
print(f"  Total BSPs: {len(bsp_names)}")
```

```
Ground truth BSPs:
  Shape original: (118000, 326)
  Shape final: (118000, 326)
  Total BSPs: 326
```

0.3 3. Cargar SAE y Extraer Features

```python
# Configuración del SAE
input_dim = 512
hidden_dim = 4096

# Construir ruta del modelo usando naming_utils
checkpoint_dir = get_checkpoint_dir(
    NAMING_META['base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos']
)
extras_id = get_extras_id(checkpoint_dir, NAMING_META['extras']) if NAMING_META.get('extras') else None

model_name = get_model_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    'best',
    extras_id=extras_id
)
model_path = checkpoint_dir / model_name

# Cargar modelo BatchTopK
sae = BatchTopKSAE(d_in=input_dim, d_sae=hidden_dim, k=NAMING_META['extras']['k']).to(device)
checkpoint = torch.load(model_path, map_location=device, weights_only=False)
sae.load_state_dict(checkpoint['model_state_dict'])
sae.eval()

print(f"SAE cargado:")
print(f"  Path: {model_path}")
print(f"  Input: {input_dim}, Hidden: {hidden_dim}, k: {NAMING_META['extras']['k']}")
print(f"  Expansion: {hidden_dim/input_dim}x")
```

```
SAE cargado:
  Path: C:\Users\Esposa\Documents\Repos\sae-othello-gpt\layer_06\models\sae_topk\sae_batch-topk_topk\1500games\sae_topk
topk_16_1500g_ex1_best.pt
  Input: 512, Hidden: 4096, k: 50
  Expansion: 8.0x
```

```python
print("Extrayendo features del SAE...")
activations_tensor = torch.from_numpy(activations).float().to(device)

with torch.no_grad():
    sae_features = sae.encode(activations_tensor)

print(f"\nFeatures SAE:")
print(f"  Shape: {sae_features.shape}")
print(f"  Device: {sae_features.device}")
print(f"  Sparsity: {(sae_features == 0).float().mean():.2%}")
print(f"  Activaciones promedio: {(sae_features > 0).sum(dim=1).float().mean():.1f}")
```

```
Extrayendo features del SAE...

Features SAE:
  Shape: torch.Size([118000, 4096])
  Device: cuda:0
  Sparsity: 98.78%
  Activaciones promedio: 50.0
```

0.4 4. Filtrar BSPs de Piezas

Coverage solo usa las 128 BSPs de piezas (mías/oponente), sin vacías.

```python
# ============================================
# CONFIGURACIÓN DE FILTRADO DE BSPs
# ============================================
USE_PLAYER   = True   # BSPs perspectiva del jugador (1 y 2)
USE_ABSOLUTE = False  # BSPs absolutas negro/blanco (B y W)
USE_STRATEGIC = False # BSPs especiales (BSP_)

# Filtrar según configuración
bsp_pieces_indices = []
bsp_pieces_names = []

for i, name in enumerate(bsp_names):
    include = False

    if USE_PLAYER and len(name) == 6 and name.startswith('BSP') and (name.endswith('1') or name.endswith('2')):
        include = True
    if USE_ABSOLUTE and (name.endswith('B') or name.endswith('W')):
        include = True
    if USE_STRATEGIC and name.startswith('BSP_'):
        include = True

    if include:
        bsp_pieces_indices.append(i)
        bsp_pieces_names.append(name)

bsp_pieces_indices = np.array(bsp_pieces_indices)
bsp_pieces_gt = bsp_ground_truth[:, bsp_pieces_indices]

# Convertir a tensor en GPU
bsp_pieces_gt_tensor = torch.from_numpy(bsp_pieces_gt).bool().to(device)

print(f"BSPs seleccionadas para Coverage:")
print(f"  Player: {USE_PLAYER} | Absolute: {USE_ABSOLUTE} | Strategic: {USE_STRATEGIC}")
print(f"  Total: {len(bsp_pieces_indices)}")
print(f"  Shape: {bsp_pieces_gt_tensor.shape}")
print(f"  Device: {bsp_pieces_gt_tensor.device}")
print(f"  Primeras 5: {', '.join(bsp_pieces_names[:5])}")
```

```
print(f"  Últimas 5: {', '.join(bsp_pieces_names[-5:])}")
# Identificador del filtrado (para naming de archivos de salida)
parts = []
if USE_PLAYER: parts.append("player")
if USE_ABSOLUTE: parts.append("absolute")
if USE_STRATEGIC: parts.append("strategic")
filter_suffix = "_".join(parts) if parts else "none"
print(f"  Filter suffix: {filter_suffix}")
```

```
BSPs seleccionadas para Coverage:
  Player: True | Absolute: False | Strategic: False
  Total: 128
  Shape: torch.Size([118000, 128])
  Device: cuda:0
  Primeras 5: BSPA11, BSPA12, BSPA21, BSPA22, BSPA31
  Últimas 5: BSPH62, BSPH71, BSPH72, BSPH81, BSPH82
  Filter suffix: player
```

0.5 5. Implementación de Coverage

0.5.1 Estrategia de optimización:

1. **Vectorización**: Procesar múltiples features en paralelo usando operaciones matriciales
2. **Batch processing**: Dividir en chunks para no saturar memoria GPU
3. **Pre-cálculo**: Calcular máximos una sola vez
4. **Early stopping**: Terminar cuando F1   1.0

```
def fast_f1_score_gpu(y_true, y_pred, eps=1e-8):
    """
    F1 score vectorizado en GPU.

    Args:
        y_true: Tensor (n_positions,) booleano
        y_pred: Tensor (n_positions, n_candidates) booleano

    Returns:
        f1_scores: Tensor (n_candidates,)
    """
    # y_true: (n_positions,) -> expand to (n_positions, 1)
    y_true_expanded = y_true.unsqueeze(1)  # (n_positions, 1)

    # Calcular TP, FP, FN
    tp = (y_true_expanded & y_pred).sum(dim=0).float()  # (n_candidates,)
    fp = (~y_true_expanded & y_pred).sum(dim=0).float()
    fn = (y_true_expanded & ~y_pred).sum(dim=0).float()

    # Precision y Recall
    precision = tp / (tp + fp + eps)
    recall = tp / (tp + fn + eps)

    # F1
    f1 = 2 * (precision * recall) / (precision + recall + eps)

    return f1


def calculate_coverage_gpu_optimized(
    sae_features,  # Tensor (n_positions, n_features) en GPU
    bsp_ground_truth,  # Tensor (n_positions, n_bsps) en GPU, bool
    thresholds=[0.0, 0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9],
    batch_size_features=512,  # Procesar 512 features a la vez
```

```python
    f_max=None,  # ← AGREGAR
    verbose=True
):
    """
    Calcula Coverage de forma optimizada para GPU.

    Procesa múltiples features y thresholds en paralelo usando operaciones matriciales.
    """
    n_positions, n_features = sae_features.shape
    n_bsps = bsp_ground_truth.shape[1]
    n_thresholds = len(thresholds)

    if verbose:
        print("="*60)
        print("CALCULATING COVERAGE (GPU OPTIMIZED)")
        print("="*60)
        print(f"Positions: {n_positions:,}")
        print(f"SAE Features: {n_features:,}")
        print(f"BSPs: {n_bsps}")
        print(f"Thresholds: {n_thresholds}")
        print(f"Batch size (features): {batch_size_features}")
        print(f"Total evaluations: {n_bsps * n_features * n_thresholds:,}")
        print("="*60)

    # Pre-calcular máximos de features
    if f_max is None:
        f_max = sae_features.max(dim=0)[0]
    active_features = (f_max > 0).nonzero(as_tuple=True)[0]
    n_active = len(active_features)

    if verbose:
        print(f"\nActive features: {n_active:,}/{n_features:,} ({n_active/n_features*100:.1f}%)")
        print(f"Reduced evaluations: {n_bsps * n_active * n_thresholds:,}")
        print()

    # Almacenar resultados
    best_f1s = torch.zeros(n_bsps, device=sae_features.device)
    best_features = torch.full((n_bsps,), -1, dtype=torch.long, device=sae_features.device)
    best_thresholds = torch.full((n_bsps,), -1.0, device=sae_features.device)

    thresholds_tensor = torch.tensor(thresholds, device=sae_features.device)

    # Procesar cada BSP
    pbar = tqdm(range(n_bsps), desc="Processing BSPs") if verbose else range(n_bsps)

    for bsp_idx in pbar:
        bsp_labels = bsp_ground_truth[:, bsp_idx]  # (n_positions,)

        # Skip si la BSP nunca está activa
        if not bsp_labels.any():
            continue

        # Procesar features activas en batches
        for batch_start in range(0, n_active, batch_size_features):
            batch_end = min(batch_start + batch_size_features, n_active)
            batch_features_idx = active_features[batch_start:batch_end]

            # Extraer features del batch
            batch_activations = sae_features[:, batch_features_idx]  # (n_positions, batch_size)
            batch_f_max = f_max[batch_features_idx]  # (batch_size,)

            # Para cada threshold, binarizar TODAS las features del batch a la vez
```

```
            for t in thresholds_tensor:
                # Binarizar: (n_positions, batch_size)
                predictions = batch_activations > (t * batch_f_max.unsqueeze(0))

                # Calcular F1 para todas las features del batch
                f1_scores = fast_f1_score_gpu(bsp_labels, predictions)  # (batch_size,)

                # Encontrar la mejor en este batch
                max_f1, max_idx_in_batch = f1_scores.max(dim=0)

                # Actualizar si es mejor que lo visto hasta ahora
                if max_f1 > best_f1s[bsp_idx]:
                    best_f1s[bsp_idx] = max_f1
                    best_features[bsp_idx] = batch_features_idx[max_idx_in_batch]
                    best_thresholds[bsp_idx] = t

            # Early stopping: si ya es casi perfecto, no revisar más batches
            if best_f1s[bsp_idx] >= 0.999:
                break

    # Calcular Coverage (macro-average)
    coverage = best_f1s.mean().item()

    # Convertir a CPU para retornar
    best_f1s_cpu = best_f1s.cpu().numpy()
    best_features_cpu = best_features.cpu().numpy()
    best_thresholds_cpu = best_thresholds.cpu().numpy()

    return coverage, best_f1s_cpu, best_features_cpu, best_thresholds_cpu

print(" Funciones de Coverage GPU definidas")
```

```
 Funciones de Coverage GPU definidas
```

0.6 6. Calcular Coverage

```
# Medir tiempo de ejecución
start_time = time.time()

coverage, best_f1s, best_features, best_thresholds = calculate_coverage_gpu_optimized(
    sae_features,
    bsp_pieces_gt_tensor,
    batch_size_features=512,  # Ajustar según memoria GPU
    verbose=True
)

elapsed_time = time.time() - start_time

print("\n" + "="*60)
print("RESULTADO COVERAGE")
print("="*60)
print(f"Coverage Score: {coverage:.4f}")
print(f"Objetivo (paper): 0.52")
print(f"Diferencia: {coverage - 0.52:+.4f}")
print(f"\nTiempo de ejecución: {elapsed_time:.2f} seg")
print(f"                      ({elapsed_time/60:.2f} min)")
print("="*60)
```

```
============================================================
CALCULATING COVERAGE (GPU OPTIMIZED)
```

```
============================================================
Positions: 118,000
SAE Features: 4,096
BSPs: 128
Thresholds: 10
Batch size (features): 512
Total evaluations: 5,242,880
============================================================

Active features: 4,094/4,096 (100.0%)
Reduced evaluations: 5,240,320

Processing BSPs: 100%|     | 128/128 [03:28<00:00,  1.63s/it]


============================================================
RESULTADO COVERAGE
============================================================
Coverage Score: 0.3331
Objetivo (paper): 0.52
Diferencia: -0.1869

Tiempo de ejecución: 209.45 seg
                      (3.49 min)
============================================================
```

0.7 7. Análisis de Resultados

```python
# Estadísticas de distribución de F1 scores
print("Distribución de F1 scores:")
print(f"  Mínimo: {best_f1s.min():.4f}")
print(f"  Máximo: {best_f1s.max():.4f}")
print(f"  Media: {best_f1s.mean():.4f}")
print(f"  Mediana: {np.median(best_f1s):.4f}")
print(f"  Std: {best_f1s.std():.4f}")
print()
print(f"BSPs con F1 > 0.8: {(best_f1s > 0.8).sum()} ({(best_f1s > 0.8).mean()*100:.1f}%)")
print(f"BSPs con F1 > 0.5: {(best_f1s > 0.5).sum()} ({(best_f1s > 0.5).mean()*100:.1f}%)")
print(f"BSPs con F1 < 0.2: {(best_f1s < 0.2).sum()} ({(best_f1s < 0.2).mean()*100:.1f}%)")
```

```
Distribución de F1 scores:
  Mínimo: 0.2399
  Máximo: 0.8464
  Media: 0.3331
  Mediana: 0.2912
  Std: 0.1070

BSPs con F1 > 0.8: 2 (1.6%)
BSPs con F1 > 0.5: 9 (7.0%)
BSPs con F1 < 0.2: 0 (0.0%)
```

```python
# Top 10 BSPs mejor detectadas
top_indices = np.argsort(best_f1s)[-10:][::-1]

print("\nTop 10 BSPs mejor detectadas:")
print("="*60)
print(f"{'BSP':<10} {'F1':<8} {'Feature':<10} {'Threshold':<10}")
```

```python
print("-"*60)
for idx in top_indices:
    bsp_name = bsp_pieces_names[idx]
    f1 = best_f1s[idx]
    feat = best_features[idx]
    thresh = best_thresholds[idx]
    print(f"{bsp_name:<10} {f1:<8.4f} {feat:<10} {thresh:<10.1f}")
```

```
Top 10 BSPs mejor detectadas:
============================================================
BSP        F1       Feature    Threshold
------------------------------------------------------------
BSPA81     0.8464   759        0.2
BSPA12     0.8301   3832       0.0
BSPH82     0.7315   488        0.0
BSPH12     0.6482   44         0.0
BSPA72     0.6060   1813       0.0
BSPH72     0.5950   465        0.0
BSPB82     0.5232   2004       0.0
BSPG71     0.5177   2288       0.0
BSPB21     0.5160   3543       0.0
BSPB71     0.4978   2906       0.0
```

```python
# Bottom 10 BSPs peor detectadas
bottom_indices = np.argsort(best_f1s)[:10]

print("\nBottom 10 BSPs peor detectadas:")
print("="*60)
print(f"{'BSP':<10} {'F1':<8} {'Feature':<10} {'Threshold':<10}")
print("-"*60)
for idx in bottom_indices:
    bsp_name = bsp_pieces_names[idx]
    f1 = best_f1s[idx]
    feat = best_features[idx]
    thresh = best_thresholds[idx]
    print(f"{bsp_name:<10} {f1:<8.4f} {feat:<10} {thresh:<10.1f}")
```

```
Bottom 10 BSPs peor detectadas:
============================================================
BSP        F1       Feature    Threshold
------------------------------------------------------------
BSPD42     0.2399   2288       0.0
BSPE52     0.2466   3626       0.0
BSPH41     0.2501   759        0.0
BSPF72     0.2511   759        0.0
BSPH71     0.2515   2288       0.0
BSPA62     0.2536   3832       0.0
BSPE61     0.2558   2288       0.0
BSPD51     0.2561   2288       0.0
BSPG52     0.2568   2288       0.0
BSPE72     0.2577   759        0.0
```

```python
# Distribución de thresholds óptimos
unique_thresholds, counts = np.unique(best_thresholds, return_counts=True)

print("\nDistribución de thresholds óptimos:")
print("="*60)
for t, count in zip(unique_thresholds, counts):
    if t >= 0:  # Ignorar -1 (BSPs sin match)
```

```
        percentage = count / len(best_thresholds) * 100
        print(f"Threshold {t:.1f}: {count:3d} BSPs ({percentage:.1f}%)")
```

```
Distribución de thresholds óptimos:
============================================================
Threshold 0.0: 127 BSPs (99.2%)
Threshold 0.2:   1 BSPs (0.8%)
```

```
# ============================================
# CONSULTA POR BSP ESPECÍFICA
# ============================================

# >>> CAMBIA AQUÍ LA BSP QUE QUIERES ANALIZAR <
#BSP_OBJETIVO = "BSPE4B"

# Buscar índice de la BSP
#bsp_idx = bsp_pieces_names.index(BSP_OBJETIVO)

# Extraer resultados ya calculados
#neurona   = best_features[bsp_idx]
#threshold = best_thresholds[bsp_idx]
#f1        = best_f1s[bsp_idx]

# f_max de la neurona ganadora (sae_features ya está en memoria)
#f_max_neurona = sae_features[:, neurona].max().item()

# Activaciones de esa neurona en todos los tableros
#activaciones_neurona = sae_features[:, neurona].cpu().numpy()

#print(f"BSP analizada:     {BSP_OBJETIVO}")
#print(f"Threshold óptimo:  {threshold:.1f}")
#print(f"F_max neurona:     {f_max_neurona:.4f}")
#print(f"Umbral efectivo:   {threshold:.1f} x {f_max_neurona:.4f} = {threshold * f_max_neurona:.4f}")
#print(f"F1 score:          {f1:.4f}")
#print(f"\nActivaciones de neurona {neurona} ({len(activaciones_neurona)} tableros):")
#for i, act in enumerate(activaciones_neurona):
 #   print(f"  Tablero {i+1:5d}: {act:.6f}")
```

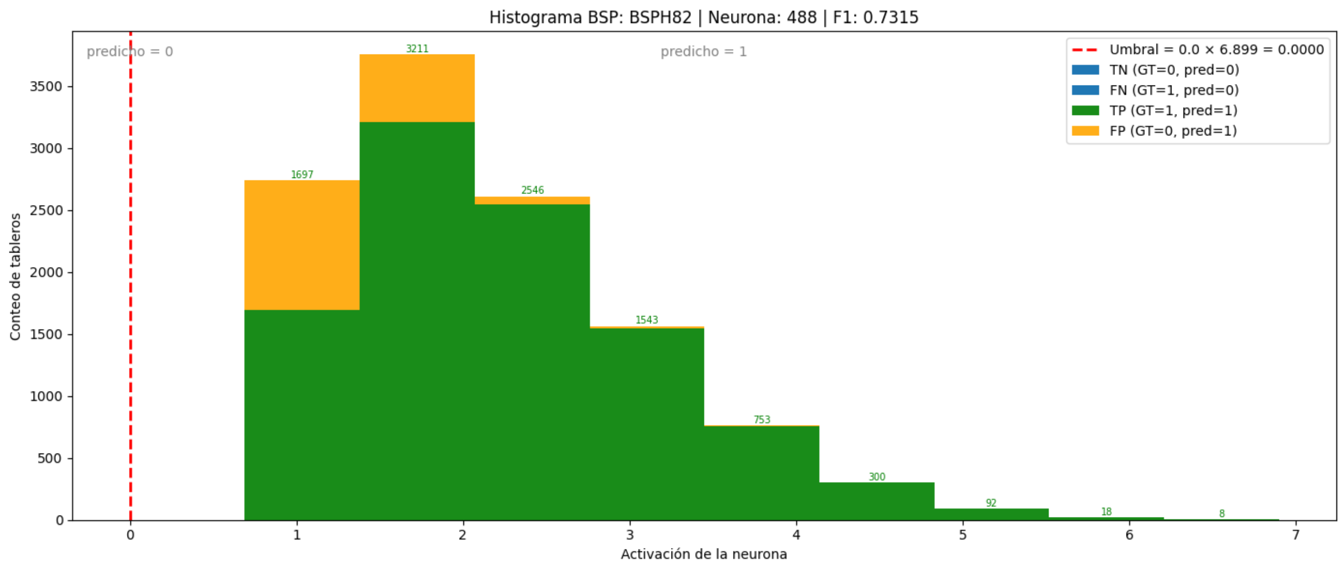
```
def plot_bsp_histogram(BSP_HISTOGRAMA, sae_features, bsp_pieces_names, bsp_pieces_gt_tensor,
                       best_features, best_thresholds, best_f1s, n_bins=20):

    # Extraer datos de la BSP
    bsp_idx_hist   = bsp_pieces_names.index(BSP_HISTOGRAMA)
    neurona_hist   = best_features[bsp_idx_hist]
    threshold_hist = best_thresholds[bsp_idx_hist]
    f1_hist        = best_f1s[bsp_idx_hist]
    f_max_hist     = sae_features[:, neurona_hist].max().item()

    # Activaciones y ground truth
    activaciones = sae_features[:, neurona_hist].cpu().numpy()
    gt           = bsp_pieces_gt_tensor[:, bsp_idx_hist].cpu().numpy()
    umbral       = threshold_hist * f_max_hist

    # Clasificar
    predicho = (activaciones > umbral).astype(int)
    tp_mask  = (gt == 1) & (predicho == 1)
    fp_mask  = (gt == 0) & (predicho == 1)
    tn_mask  = (gt == 0) & (predicho == 0)
    fn_mask  = (gt == 1) & (predicho == 0)
```

```python
    # Bins
    bins        = np.linspace(0, activaciones.max(), n_bins + 1)
    bin_centers = (bins[:-1] + bins[1:]) / 2
    bin_width   = bins[1] - bins[0]

    tp_counts, _ = np.histogram(activaciones[tp_mask], bins=bins)
    fp_counts, _ = np.histogram(activaciones[fp_mask], bins=bins)
    tn_counts, _ = np.histogram(activaciones[tn_mask], bins=bins)
    fn_counts, _ = np.histogram(activaciones[fn_mask], bins=bins)

    left_mask  = bin_centers <= umbral
    right_mask = bin_centers > umbral

    # Graficar
    fig, ax = plt.subplots(figsize=(14, 6))

    ax.bar(bin_centers[left_mask], tn_counts[left_mask],
           width=bin_width, color='steelblue', alpha=0.9,
           label='TN (GT=0, pred=0)', align='center')
    ax.bar(bin_centers[left_mask], fn_counts[left_mask],
           width=bin_width, bottom=tn_counts[left_mask],
           color='lightcoral', alpha=0.9,
           label='FN (GT=1, pred=0)', align='center')
    ax.bar(bin_centers[right_mask], tp_counts[right_mask],
           width=bin_width, color='green', alpha=0.9,
           label='TP (GT=1, pred=1)', align='center')
    ax.bar(bin_centers[right_mask], fp_counts[right_mask],
           width=bin_width, bottom=tp_counts[right_mask],
           color='orange', alpha=0.9,
           label='FP (GT=0, pred=1)', align='center')

    # Números TN
    for i, x in enumerate(bin_centers[left_mask]):
        total = tn_counts[left_mask][i] + fn_counts[left_mask][i]
        if tn_counts[left_mask][i] > 0:
            ax.text(x, total + 0.3, str(tn_counts[left_mask][i]),
                    ha='center', va='bottom', fontsize=7, color='steelblue')

    # Números TP
    for i, x in enumerate(bin_centers[right_mask]):
        total = tp_counts[right_mask][i] + fp_counts[right_mask][i]
        if tp_counts[right_mask][i] > 0:
            ax.text(x, total + 0.3, str(tp_counts[right_mask][i]),
                    ha='center', va='bottom', fontsize=7, color='green')

    # Umbral
    ax.axvline(x=umbral, color='red', linewidth=2, linestyle='--',
               label=f'Umbral = {threshold_hist:.1f} × {f_max_hist:.3f} = {umbral:.4f}')

    ax.set_xlabel('Activación de la neurona')
    ax.set_ylabel('Conteo de tableros')
    ax.set_title(f'Histograma BSP: {BSP_HISTOGRAMA} | Neurona: {neurona_hist} | F1: {f1_hist:.4f}')
    ax.legend()
    ax.text(umbral * 0.5, ax.get_ylim()[1] * 0.95, 'predicho = 0',
            ha='center', fontsize=10, color='gray')
    ax.text(umbral + (activaciones.max() - umbral) * 0.5, ax.get_ylim()[1] * 0.95, 'predicho = 1',
            ha='center', fontsize=10, color='gray')

    plt.tight_layout()
    plt.show()

    print(f"\nResumen:")
```

```
    print(f"  TP: {tp_mask.sum()} | FP: {fp_mask.sum()}")
    print(f"  TN: {tn_mask.sum()} | FN: {fn_mask.sum()}")
    print(f"  F1: {f1_hist:.4f}")

print("  Función plot_bsp_histogram definida")
```

```
  Función plot_bsp_histogram definida
```

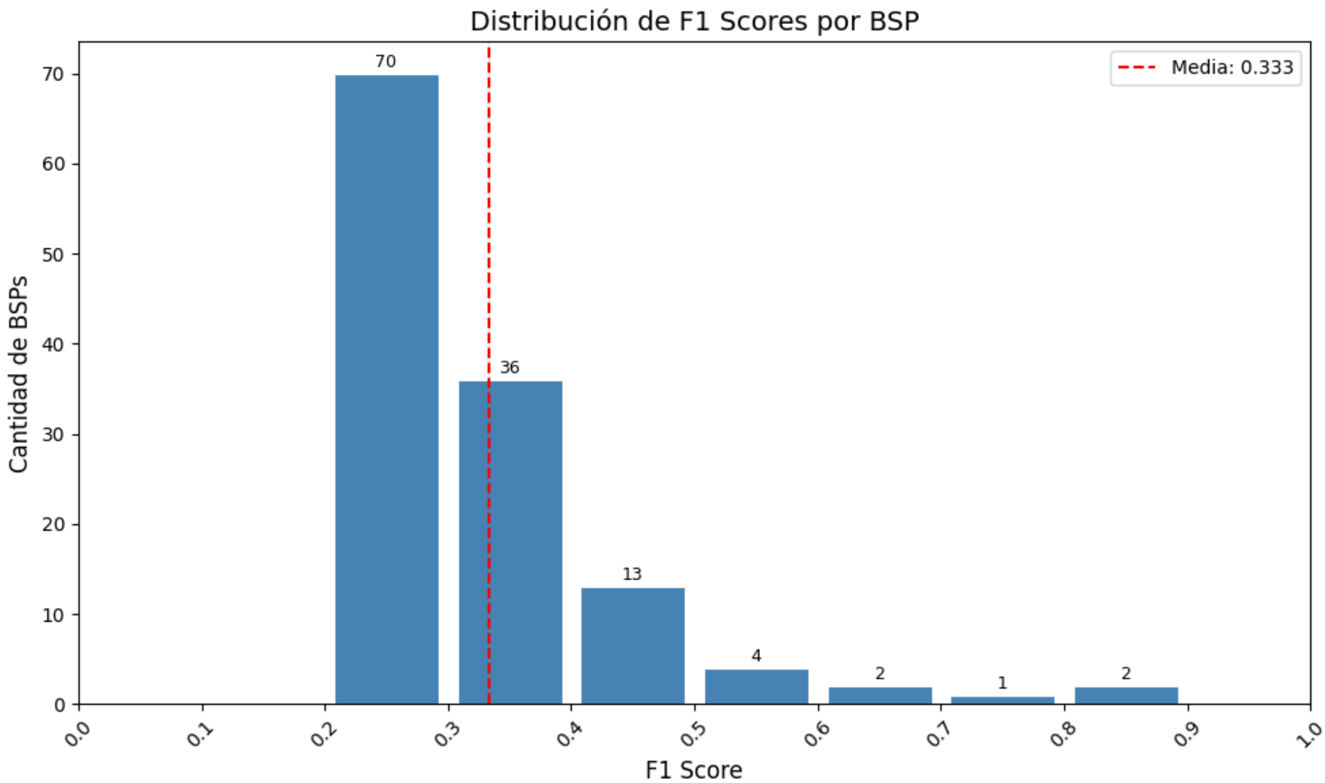
```
# >>> CAMBIA AQUÍ LA BSP QUE QUIERES VISUALIZAR <
import matplotlib.pyplot as plt
plot_bsp_histogram(
    BSP_HISTOGRAMA        = "BSPH82",
    sae_features          = sae_features,
    bsp_pieces_names      = bsp_pieces_names,
    bsp_pieces_gt_tensor = bsp_pieces_gt_tensor,
    best_features         = best_features,
    best_thresholds       = best_thresholds,
    best_f1s              = best_f1s,
    n_bins                = 10
)
```




```
Resumen:
  TP: 10168 | FP: 1679
  TN: 100368 | FN: 5785
  F1: 0.7315
```

```
import matplotlib.pyplot as plt
import numpy as np

def plot_f1_distribution(best_f1s, bsp_pieces_names, n_bins=10):
    """
    Gráfico 1: Distribución de F1 scores por BSP.

    Args:
        best_f1s: array numpy con el mejor F1 score de cada BSP
        bsp_pieces_names: lista con los nombres de las BSPs
        n_bins: número de bins del histograma
    """
    fig, ax = plt.subplots(figsize=(10, 6))
```

```python
    bins = np.linspace(0, 1, n_bins + 1)
    counts, edges = np.histogram(best_f1s, bins=bins)
    bin_centers = (edges[:-1] + edges[1:]) / 2
    bin_width = edges[1] - edges[0]

    bars = ax.bar(
        bin_centers, counts,
        width=bin_width * 0.85,
        color='steelblue',
        edgecolor='white',
        linewidth=0.8
    )

    # Etiquetas encima de cada barra
    for bar, count in zip(bars, counts):
        if count > 0:
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + 0.3,
                str(count),
                ha='center', va='bottom', fontsize=9
            )

    ax.set_xlabel('F1 Score', fontsize=12)
    ax.set_ylabel('Cantidad de BSPs', fontsize=12)
    ax.set_title('Distribución de F1 Scores por BSP', fontsize=14)
    ax.set_xticks(bins)
    ax.set_xticklabels([f'{b:.1f}' for b in bins], rotation=45)
    ax.set_xlim(0, 1)

    # Línea de la media
    mean_f1 = best_f1s.mean()
    ax.axvline(mean_f1, color='red', linestyle='--', linewidth=1.5, label=f'Media: {mean_f1:.3f}')
    ax.legend(fontsize=10)

    plt.tight_layout()
    plt.show()

    print(f"Total BSPs: {len(best_f1s)}")
    print(f"Media F1: {mean_f1:.4f}")
    print(f"BSPs con F1 = 0: {(best_f1s == 0).sum()}")
    print(f"BSPs con F1 > 0.5: {(best_f1s > 0.5).sum()}")
    print(f"BSPs con F1 > 0.9: {(best_f1s > 0.9).sum()}")


# Ejecutar
plot_f1_distribution(best_f1s, bsp_pieces_names)
```

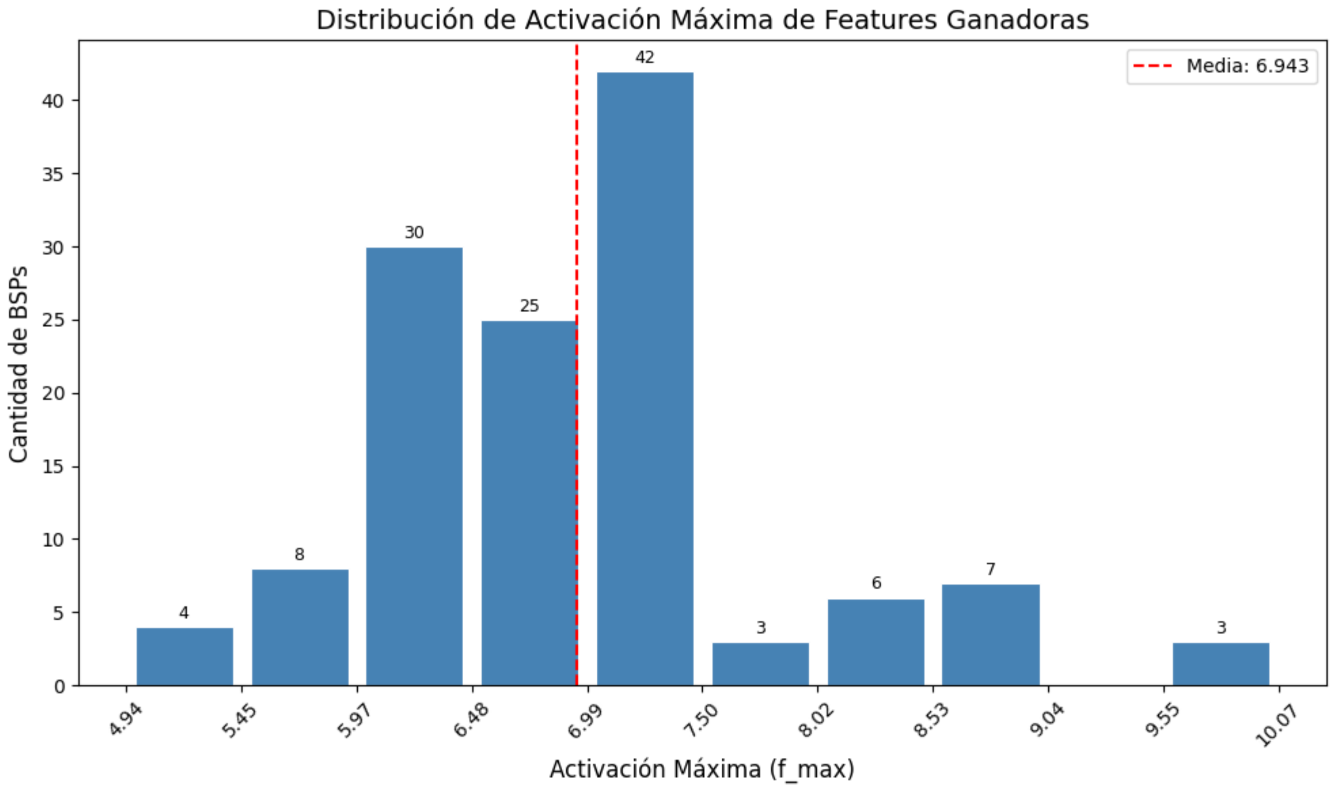



```
Total BSPs: 128
Media F1: 0.3331
BSPs con F1 = 0: 0
BSPs con F1 > 0.5: 9
BSPs con F1 > 0.9: 0
```

```python
# Extraer f_max de las features activas
f_max = sae_features.max(dim=0)[0]  # (n_features,)
```

```python
def plot_max_activation_distribution(best_features, f_max, bsp_pieces_names, n_bins=10):
    """
    Gráfico 2: Distribución de activación máxima de la feature ganadora por BSP.

    Args:
        best_features: array numpy con el índice de la feature ganadora de cada BSP
        f_max: tensor con la activación máxima de cada feature
        bsp_pieces_names: lista con los nombres de las BSPs
        n_bins: número de bins del histograma
    """
    # Obtener f_max de la feature ganadora de cada BSP
    # Ignorar BSPs sin feature ganadora (best_features == -1)
    valid_mask = best_features >= 0
    valid_features = best_features[valid_mask]
    f_max_winners = f_max[valid_features].cpu().numpy()

    fig, ax = plt.subplots(figsize=(10, 6))

    bins = np.linspace(f_max_winners.min(), f_max_winners.max(), n_bins + 1)
    counts, edges = np.histogram(f_max_winners, bins=bins)
    bin_centers = (edges[:-1] + edges[1:]) / 2
    bin_width = edges[1] - edges[0]

    bars = ax.bar(
```

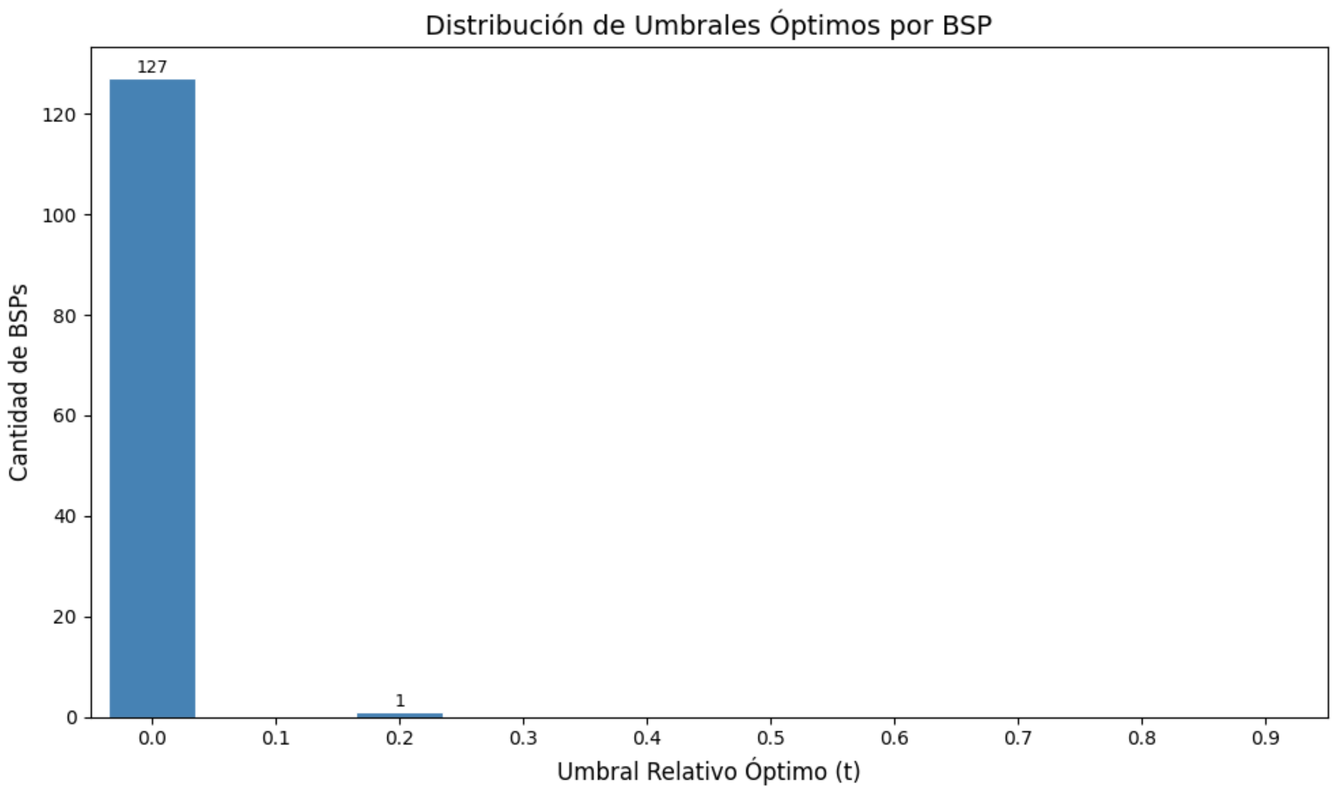
```
        bin_centers, counts,
        width=bin_width * 0.85,
        color='steelblue',
        edgecolor='white',
        linewidth=0.8
    )

    # Etiquetas encima de cada barra
    for bar, count in zip(bars, counts):
        if count > 0:
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + 0.3,
                str(count),
                ha='center', va='bottom', fontsize=9
            )

    # Línea de la media
    mean_fmax = f_max_winners.mean()
    ax.axvline(mean_fmax, color='red', linestyle='--', linewidth=1.5,
               label=f'Media: {mean_fmax:.3f}')
    ax.legend(fontsize=10)

    ax.set_xlabel('Activación Máxima (f_max)', fontsize=12)
    ax.set_ylabel('Cantidad de BSPs', fontsize=12)
    ax.set_title('Distribución de Activación Máxima de Features Ganadoras', fontsize=14)
    ax.set_xticks(edges)
    ax.set_xticklabels([f'{b:.2f}' for b in edges], rotation=45)

    plt.tight_layout()
    plt.show()

    print(f"Total BSPs con feature ganadora: {valid_mask.sum()}")
    print(f"BSPs sin feature ganadora: {(~valid_mask).sum()}")
    print(f"Media f_max ganadora: {mean_fmax:.4f}")
    print(f"Mínimo f_max: {f_max_winners.min():.4f}")
    print(f"Máximo f_max: {f_max_winners.max():.4f}")


# Ejecutar
plot_max_activation_distribution(best_features, f_max, bsp_pieces_names)
```




```
Total BSPs con feature ganadora: 128
BSPs sin feature ganadora: 0
Media f_max ganadora: 6.9425
Mínimo f_max: 4.9416
Máximo f_max: 10.0665
```

```python
def plot_threshold_distribution(best_thresholds, bsp_pieces_names):
    """
    Gráfico 3: Distribución de umbrales óptimos por BSP.

    Args:
        best_thresholds: array numpy con el umbral óptimo de cada BSP
        bsp_pieces_names: lista con los nombres de las BSPs
    """
    # Ignorar BSPs sin feature ganadora (best_thresholds == -1)
    valid_mask = best_thresholds >= 0
    valid_thresholds = best_thresholds[valid_mask]

    fig, ax = plt.subplots(figsize=(10, 6))

    # Los umbrales son discretos: 0.0, 0.1, 0.2, ..., 0.9
    threshold_values = np.arange(0.0, 1.0, 0.1)
    counts = [(np.abs(valid_thresholds - t) < 0.05).sum() for t in threshold_values]

    bars = ax.bar(
        threshold_values, counts,
        width=0.07,
        color='steelblue',
        edgecolor='white',
        linewidth=0.8
    )

    # Etiquetas encima de cada barra
```

```python
    for bar, count in zip(bars, counts):
        if count > 0:
            ax.text(
                bar.get_x() + bar.get_width() / 2,
                bar.get_height() + 0.3,
                str(count),
                ha='center', va='bottom', fontsize=9
            )

    ax.set_xlabel('Umbral Relativo Óptimo (t)', fontsize=12)
    ax.set_ylabel('Cantidad de BSPs', fontsize=12)
    ax.set_title('Distribución de Umbrales Óptimos por BSP', fontsize=14)
    ax.set_xticks(threshold_values)
    ax.set_xticklabels([f'{t:.1f}' for t in threshold_values])
    ax.set_xlim(-0.05, 0.95)

    plt.tight_layout()
    plt.show()

    print(f"Total BSPs con umbral óptimo: {valid_mask.sum()}")
    print(f"BSPs sin feature ganadora: {(~valid_mask).sum()}")
    print(f"Umbral más frecuente: {threshold_values[np.argmax(counts)]:.1f}")


# Ejecutar
plot_threshold_distribution(best_thresholds, bsp_pieces_names)
```




```
Total BSPs con umbral óptimo: 128
BSPs sin feature ganadora: 0
Umbral más frecuente: 0.0
```

0.8 8. Guardar Resultados

```
# Guardar resultados en el directorio de métricas del experimento
results = {
    'coverage': coverage,
    'best_f1s': best_f1s,
    'best_features': best_features,
    'best_thresholds': best_thresholds,
    'bsp_names': bsp_pieces_names,
    'execution_time_seconds': elapsed_time
}

metrics_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)
output_path = metrics_dir / f"coverage_{filter_suffix}.npz"
np.savez(output_path, **results)

print(f" Resultados guardados en:")
print(f"  {output_path}")
```

```
 Resultados guardados en:
  c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_batch-topk_topk\1500games\ex1\met
```

0.9 9. Generar PDF con Quarto

```
import subprocess

# Guardar PDF en el mismo directorio que el .npz (metrics/)
pdf_dir = get_metrics_dir(
    NAMING_META['experiment_base_path'],
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    extras_id=extras_id
)

pdf_name = get_report_name(
    NAMING_META['variante'],
    NAMING_META['tecnica'],
    NAMING_META['capa'],
    NAMING_META['juegos'],
    f'coverage_{filter_suffix}',
    extras_id=extras_id
)

notebook_name = 'coverage_sae_batchtopk.ipynb'
output_path = pdf_dir / pdf_name

print(f' Generando PDF con Quarto...')
print(f' Archivo de salida: {output_path}')

result = subprocess.run(
```

```
    f'quarto render {notebook_name} --to pdf --output-dir "{pdf_dir}" --output {pdf_name}',
    shell=True,
    capture_output=True,
    text=True
)

if result.returncode == 0:
    print(f' PDF generado exitosamente: {output_path}')
else:
    print(f' Error al generar PDF:')
    print(result.stderr)
```

```
 Generando PDF con Quarto...
 Archivo de salida: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_batch-topk_topk\1500games\ex1\metrics\sae_topk_batch-topk_l6_1500g_ex1_coverage_player.pdf
 PDF generado exitosamente: c:\Users\Esposa\Documents\Repos\othello_world\sae\experiments\layer_06\sae_topk\sae_batch-topk_topk\1500games\ex1\metrics\sae_topk_batch-topk_l6_1500g_ex1_coverage_player.pdf
```